



DFSMSHsm

Product overview

DFSMSHsm performs two main functions, Availability Management and Space Management. Availability Management is used to make data available by automatically copying new and changed data set to backup volumes. Space Management is used to manage DASD space by enabling inactive data sets to be moved off fast-access storage devices thus creating free space for new allocations. DFSMSHsm also provides for other supporting functions that are essential to your installations DFSMSHsm environment.

The following list illustrates the functions performed by DFSMSHsm for Availability Management, Space Management and these supporting functions.

Main Functions

↓ Availability Management

↓ ABARS

↓ Backup

↓ Dump

↓ Recover

↓ Space Management

↓ Expiration

↓ Extent Reduction

↓ Interval Migration

↓ Migration

↓ Recall

↓ Recycle

↓ Primary Space Management

↓ Secondary Space Management

Supporting Functions

↓ CDS (Backup, Journaling, and Monitoring)

↓ Logging and Diagnosis

↓ MVS Environment for DFSMSHsm

↓ Security

↓ Tape Environment

A Complementary Solution

↓ DFSMS Optimizer

Availability Management

Availability management ensures that a recent copy of your DASD data sets exists. The purpose is to ensure that lost or damaged data sets can be retrieved at the correct level. To do this, availability management should automatically and periodically:

- copy data sets from DASD to tape volumes
- copy changed data sets on DASD volumes either to other DASD or tape volumes
- copy data sets of an application to tape volumes so they can be taken to another computer site for recovery

By performing these functions, availability management can help:

- guard against users accidentally losing or incorrectly changing data sets
- recover data after a volume is lost due to a hardware failure
- protect against loss of data due to a disaster
- ensure data which must be kept due to external requirements is retained (e.g. tax records)

ABARS

ABARS facilitates a point-in-time backup of a collection of related data in a consistent manner. This group of related data is defined to ABARS as an aggregate. An aggregate is user-defined and is usually a collection of data sets that have some sort of relationship, such as all the specific data sets required for an application (e.g., payroll). ABARS backs up the data directly from DASD (either SMS or non-SMS managed), DFSMSHsm-owned ML1 DASD, or DFSMSHsm-owned ML2 tape without the need for intermediate staging capacity. The backup copies are created in a device-independent format. ABARS is generally used for disaster recovery or to move applications across non-sharing systems. At the recovery site, ABARS is used to recover the data sets as well as allocating user-defined empty data sets that the customer requires for operation of his environment.

ABARS has three components:

- **Aggregate group definition:** The aggregate group lists the selection data set names, instruction data set name, and the management class for the aggregate. This component is implemented for the Interactive Storage Management facility (ISMF) and requires the storage management subsystem (SMS) address space to be active.
- **Aggregate backup (ABACKUP):** The ABACKUP command backs up a user-defined group (aggregate group) of data sets at that moment in time. This DFSMSHsm function also requires the SMS address space to be active.
- **Aggregate recover (ARECOVER):** The ARECOVER command recovers data sets that were previously backed up by an aggregate backup. The SMS address space is not required, but it is highly recommended to simplify the recovery process. This address space is required to recover VSAM data sets in the ALLOCATE list and any extended format data sets such as partitioned data set extended (PDSE).

IBM also provides the following software products from Mainstar Software Corporation to enhance your ABARS environment:

- The Mainstar Disaster Recovery Utilities provide additions functions that complement and extend those in the DFSMSHsm ABARS solution.

➡ *DFSMSHsm ABARS and Mainstar Solutions*, SG24-5089-00 is a new Redbook that discusses the ABARS function of DFSMSHsm and two Mainstar products ASAP and ABARS Manager.

Backup

Backup is the process of copying a data set from a level 0 or an ML1 volume to a daily backup volume. This copy is called a backup version. The purpose of backup is to have copies of data sets in case something happens to the original data sets.

DFSMSHsm can create backup versions of data sets either automatically, by command, or within a job stream. Backup creates a logical copy of the data set and is not device dependent.

DFSMSHsm automatically creates backup versions of data sets on specified days beginning at a specified time of day. The data sets must meet eligibility criteria and must be on DFSMSHsm-managed volumes that have been designated for automatic backup.

You can use DFSMSHsm commands to backup up specific data sets or volumes. The BACKDS command creates a backup version of a specific data set. The BACKVOL command creates a backup version of all eligible data sets from a specific level 0 volume or backup the DFSMSHsm control data sets and the journal.

Dump

Dump is the process of copying all data from a DASD volume to dump tape volumes. Full-volume dump is an extension of DFSMSHsm's availability management that invokes DFSMSDss through the DFSMSDss application interface. Full-volume dump backs up the

entire allocated space of DFSMSHsm-managed DASD volumes and ML1 volumes either automatically or by command. Non-DFSMSHsm-managed volumes are dumped only by command. The purpose of the full-volume dump is to expedite the recovery process when an entire volume is lost or damaged and to supplement the incremental recovery process. A dump copy of a volume is a physical copy and is device dependent.

Recover

Recovery and restore are processes that are requested only by command, not automatically, for backed up data sets. Recovery is the process of retrieving a full-volume dump or a backup version of a data set or a volume. Restore signifies that DFSMSdss is used to retrieve dumped data from dump volumes. You can use restore or recover processing to:

- Recover a data set that has been lost or damaged
- Recover an earlier version of the data set without deleting the current version
- Restore a volume from a full-volume dump and update the volume from later incremental backup versions
- Restore a data set from a dump copy
- Restore a volume from a full-volume dump
- Recover a volume from DFSMSHsm backup versions

DFSMSHsm volume recovery can use incremental backups or full-volume dumps or both. One DFSMSHsm RECOVER command can be used to request both a volume restore and an incremental volume recovery.

Space Management

Space management allows you to keep DASD space available for users in order to meet the service-level objectives for your system. The purpose of space management is to manage your DASD storage efficiently. To do this, it automatically and periodically:

- Expires data sets that have passed their expiration dates
- Migrates unused data sets to locations where they can be stored less expensively or in a form that saves space on the volume
- Reduces the space occupied by data by freeing the unused portions of the data set (release)

Expiration

During automatic primary or secondary space management, DFSMSHsm deletes data sets that have an explicit or implied expiration date. If the data set has no explicit expiration date, it is controlled by the management class attributes. The storage administrator can control whether DFSMSHsm deletes expired data sets having explicit expiration dates.

Extent Reduction

During automatic primary space management, DFSMSHsm migrates the data sets that are candidates for extent reduction and schedules a recall for those same data sets. During the migration, DFSMSHsm moves only the valid data from the data sets. Data sets are then recalled into fewer extents and are better able to be extended during normal processing.

Interval Migration

Automatic interval migration is an option that invokes migration when DFSMSHsm-managed volumes become full during high-activity periods. If the storage administrator chooses this option, DFSMSHsm automatically checks the level-of-occupancy of all DFSMSHsm-managed volumes periodically. If any volume's level of occupancy exceeds a given threshold, DFSMSHsm automatically performs a subset of the space management functions on the volume. The threshold you select should be one that would be exceeded only when your installation's activity exceeds its usual peak. For those volumes requiring

interval migration, DFSMSHsm can process up to 15 volume migration tasks concurrently.

During automatic interval migration on a volume, the expired data sets are deleted, then the largest eligible data sets are moved first so that the level of occupancy threshold can be reached sooner. Data sets are not migrated from ML1 to ML2 volumes during interval migration.

For system-managed volumes, DFSMSHsm uses the level-of-occupancy thresholds established for the storage group to control this function. For non-system-managed volumes, DFSMSHsm uses the level-of-occupancy thresholds established for the DFSMSHsm-managed volume to control this function.

Tape Mount Management (TMM) is a methodology used to optimize tape subsystem operation and use, consisting of hardware and software facilities used to manage tape data efficiently. By using DFSMSHsm in conjunction with your ACS routines to implement the TMM methodology, you can write multiple output data sets to a single tape, with a single tape mount. This not only improves tape use, but also greatly reduces the number of tape mounts required by your installation.

The following table illustrates the differences between normal interval migration processing and TMM processing:

DFSMSHsm Options / Controls	Interval Migration	TMM with SMS
Hourly Option Note: Hourly migration is not performed on a host that is currently running Primary Space Management.	Storage Group having AM=Y Note: Specifying AM=Y for a Storage Group requests that Primary Space Management and interval migration both be performed on volumes in this Storage Group. An option of AM=P exists if only Primary Space Management is desired. Non-SMS volumes also perform this way.	Storage Group having AM=I Note: This option of AM=I is referred to as Tape Mount Management (TMM).
SETSYS INTERVALMIGRATION	Volumes meeting trigger threshold may be processed hourly on this host.	N/A
SETSYS NOINTERVALMIGRATION	Volumes are not processed hourly on this host.	N/A
Threshold Range	1-99	0-99
Trigger Threshold	High threshold	Average of high and low threshold
Max interval tasks of 0	No hourly migration will be done on this host. Note: The option of NOINTERVALMIGRATION provides the same capability for this category of data.	No hourly migration will be done on this host. Note: Be aware that if you set the SETSYS MAXINTERVALTASKS parameter to zero, you are preventing hourly migration for all Storage Groups and non-SMS volumes on this host.
Max interval tasks not 0	'x' tasks available to do hourly migration of volumes meeting trigger threshold	'x' tasks available to do hourly migrations of volumes meeting trigger

	trigger threshold.	threshold.
AMHOST = blank	Hourly migration can be performed on any host permitting interval migration and having interval tasks.	Hourly migration can be performed on any host having interval tasks.
AMHOST = host or sysplex name Note: If you want to allow hourly processing on more than one host but not on all hosts of a sysplex, then you can specify the combination of a sysplex restricting name but specify SETSYS MAXINTERVALTASKS(0) on the hosts where you don't want to permit hourly migration.	Primary Space Management and hourly migration limited to this host(s).	Primary Space Management and hourly migration limited to this host(s).

Migration

In DFSMSHsm, migration is the process of moving a cataloged data set from a:

- DFSMSHsm-managed volume to a migration level 1 or a migration level 2 volume (Primary Space migration)
- Migration level 1 volume to a migration level 2 volume (Secondary Space migration)
- Volume not managed by DFSMSHsm to a migration level 1 or migration level 2 volume

Recall

The recall process returns a migrated data set from a ML1 or ML2 volume to a DFSMSHsm-managed volume. When a user refers to the data set, DFSMSHsm reads the system catalog for the volume serial number. If the volume serial number is MIGRAT, DFSMSHsm finds the migrated data set, recalls it to a DFSMSHsm-managed volume, and updates the catalog. The result of the recall process is a data set that resides on a user volume in a user-readable format. The recall can also be requested by a DFSMSHsm command.

Recycle

Data on tapes is invalidated over time by the expiration of migrated data sets or the generation of more recent backup data sets. DFSMSHsm provides the capability of recycling backup or migration tapes when they contain less than an installation-defined percentage of valid data. Recycling transfers the valid data sets from the tapes being recycled and consolidates the data on another tape, thus making one tape with all unexpired data sets and leaving the recycled tapes available for reuse.

Primary Space Management

Primary space management deletes expired migrated data sets and migrates eligible data sets from DFSMSHsm-managed volumes. The storage administrator specifies the day in the cycle and the time of day automatic primary space management is to be performed.

Secondary Space Management

Secondary space management deletes expired migrated data sets and migrates eligible data sets from ML1 volumes to ML2 volumes. The storage administrator specifies the day in the cycle and the time of day automatic secondary space management is to be

performed.

Supporting Functions

Control Data Sets (CDS)

DFSMSHsm control data sets are system-type data sets that DFSMSHsm uses to keep track of all DFSMSHsm-owned data. They consist of migration, backup, and offline control data sets. The control data sets are an inventory of low-activity and inactive data that has been stored by DFSMSHsm, and they are used by DFSMSHsm to manage its environment. DFSMSHsm logs its transactions and maintains multiple backup versions of its CDSs for recovery purposes as specified by the user. The journal, a record of all transactions posted to the control data sets between backups, is used in the recovery process to update the recovered control data sets to the point-of-failure.

The control data sets can be accessed in record-level sharing (RLS) mode. Accessing the control data sets in RLS mode allows DFSMSHsm to exploit parallel sysplex technology and to benefit from the serialization features of the coupling facility when performing functions on two or more processors.

Logging and Diagnosis

During DFSMSHsm processing, the problem determination aid (PDA) facility gathers diagnostic information about DFSMSHsm's processing, stores this information in a circular file within storage, and periodically writes it to a circular file on DASD. This circular type of file appends data until full; then, starting at the beginning of the file, subsequent incoming data overwrites the data already there.

The circular DASD file consists of two data sets, ARCPDOX and ARCPDOY. ARCPDOX is the active data set. You can make ARCPDOY become the active data set, and the ARCPDOX becomes the inactive data set, by issuing the SWAPLOG PDA command. This allows you to take the active data set and make it inactive for the purpose of off-loading the diagnostic information that has been gathered.

MVS Environment for DFSMSHsm

DFSMSHsm provides for options that allow you to customize your MVS environment for DFSMSHsm processing. You can choose from many SETSYS options and you can use installation exits to customize DFSMSHsm processing.

DFSMS installation exits allow you to extend the capabilities of DFSMS to customize your installation's MVS/ESA operation. The DFSMSHsm installation exits can be used to customize DFSMSHsm processing. The DFSMSHsm installation exits fall into two categories: exits that support basic DFSMSHsm functions and exits that support DFSMSHsm ABARS functions. You can refer to the *DFSMS Installation Exits*, SC26-4908, for more information about these exits.

Security

DFSMSHsm protects data sets from unauthorized access by controlling access to data sets and tasks. Access can be controlled by the use of both passwords and security programs at the same time. However, system-managed data sets are not password protected. If a data set is password protected and security-program protected, DFSMSHsm allows access to the data set without checking the password if the security program authorizes the access.

DFSMSHsm provides three basic methods for protecting tape volumes under its control: RACF, expiration date, and passwords. You can choose more than one tape protection option.

Because DFSMSHsm operates as an MVS-authorized task, it can manage data sets automatically regardless of their password or security-program protection. To prevent

unwanted changes of the parameters that control all data sets, DFSMSHsm has classified it commands as authorized and nonauthorized. Authorized commands can be entered only by a user specifically authorized by the storage administrator. Generally, authorized commands can affect data sets not owned by the person issuing the command. Nonauthorized commands can be entered by any user, but they generally affect only those data sets for which the user has appropriate access authority.

Authorized users are assigned either USER or CONTROL authorization. Users with USER authorization can enter any DFSMSHsm command except the one to authorize other users. Users with CONTROL authorization can enter any DFSMSHsm command.

Tape Environment

DFSMSHsm works with all major tape management systems. DFSMSHsm provides an internal interface to IBM's tape management system, DFSMSrmm, and provides a general exit as an interface to other tape management systems.

DFSMSHsm is generally used with a global or public scratch pool. That is, tapes enter DFSMSHsm's usage by a non-specific tape mount and then when all data has been invalidated, the tape is returned to the scratch pool. DFSMSHsm is also able to work with a private scratch pool when tapes are specifically identified for the private use of DFSMSHsm.

Tapes can be protected by RACF or expiration dates and are expected to be included in a tape management product of the customers choice.

DFSMSHsm takes advantage of the single-file format and the compaction algorithms available on cartridge-type devices, such as the IBM 3480, 3490, 3490E, and 3590-1. The single-file format allows for better performance and utilization of cartridges since they can contain hundreds or thousands of data sets on a single tape cartridge. DFSMSHsm uses the capacity of the cartridges and provides ways to handle contentions for different data sets residing on the same cartridge.

DFSMSHsm provides tape duplication for migration or backup tapes by providing a second or alternate tape copy. This duplicate tape can be made by concurrent creation (duplex) or by copying when the tape becomes full (TAPECOPY).

Physical volume dumps can have up to 5 copies made concurrently and the dumps from multiple DASD's can be written to a single Magstar cartridge.

Monitor and Automate DFSMSHsm using DFSMS Optimizer

IBM's DFSMS Optimizer provides performance analysis and Management Class analysis capability enabling you to understand how you are managing data today, simulate the impact of change, and analyze your data management costs. The DFSMSHsm Monitor can be used for real-time monitoring and automation of DFSMSHsm functions.

The DFSMS Optimizer Management Class analyzer can be used to optimize the choice of migration attributes to reduce costs and migration/recall thrashing by management class. Thrashing is defined as: an overcommitment of resources which results in DFSMSHsm's inability to complete work in the time allocated. The overhead can be self-generated (requesting too many actions during a period of time), or externally generated. An example of externally generated overhead occurs when you define a management class to migrate every day when the data sets are still being used. This condition forces recalls, which are then migrated, and becomes thrashing.